

wish has come true in the form of two very promising new programming languages—Vala and Genie—which are going to solve our problem for us, as we see in this article.

I used Puppy Linux 4.2.1 and the Ubuntu 9.10 64-bit desktop edition to test the code I present in the article, and to generate screenshots.

Platform and libraries

Before beginning with Vala and Genie, let's get familiar with the platform and dependencies expected with these languages. Vala and Genie use the GLib and GObject libraries, which make it particularly easy to write GTK code. GLib provides a portable wrapper for many lower-level functions and data types, as well as various advanced data structures; it is the C++ STL counterpart in C in most of the cases. GObject is the type and object system that provides portable object orientation and automatic memory management for objects, through reference counting. For more detailed information about GLib and GObject, see the link provided in the *References* section.

The Vala/Genie compiler even has a command-line option `--profile=posix`, to eliminate the need for GLib/GObject libraries. Though that means sacrificing some functionality, it enables these powerful and easy languages to be used as a replacement for C in areas like embedded systems or boot-disks, where tiny static executables are required. You can create very tiny self-contained native executables programmed in Vala or Genie, by using alternate light versions of *libc* and performing static linking.

Figure 1 shows how you get an executable from Vala and Genie source code. The build process first converts Vala and Genie source code to a GObject-based C-code representation, then invokes the C compiler to generate a host-platform-specific executable. In a departure from the prevalent *one compiler, one syntax* standard, the Vala compiler converts two different syntax forms into an intermediate form—Vala's syntax, inspired by C#, Java, etc; and Genie's, inspired by Python, Boo and the like. This makes choosing between Vala or Genie a matter of preference.

The GNOME libraries make available a rich set of functionality to both languages -- from GUI to multimedia. Besides these, there are language bindings available to several third-party libraries for Vala and Genie.

To work with these languages, the normal prerequisites are GNOME development libraries and C runtime libraries—and those are guaranteed to be available wherever a GNOME development environment is available. You don't need any extra performance-dampening runtime components like virtual machines or extra runtime libraries, as is the case for C#, Java, Python, etc. This hybrid development method provides the productivity and flexibility of managed languages, with the runtime performance of compiled languages—for both languages, runtime performance is close to or equal to that of C programs.

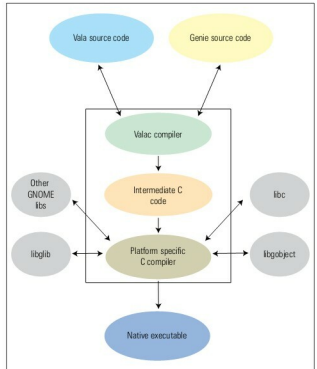


Figure 1: Executable generation from Vala and Genie

Installing the Vala and Genie compiler

To compile source code, we need the Vala compiler, *valac*. If you don't insist on having the latest version, then you can install *valac* on Ubuntu with the following command line:

```
sudo apt-get install valac
```

To get the latest version, you'll have to build and install it from the source tarball, since both the languages and compiler are in continuous development. You need *bison*, *flex*, *gcc* and *glib* sources installed to build *valac*. If you are using Puppy Linux version 4.3, then Vala is already available in the "devx" SFS file, which provides all you need for a C/C++/Vala/Genie programming environment. On Ubuntu, issue the following command in a terminal:

```
sudo apt-get install gcc libglib2.0-dev bison flex
```

Now download the latest Vala compiler source tarball from the Vala homepage (see the *References* section for the URL.) and issue the following commands in a terminal to build and install *valac*:

```
tar -jxvf Vala-version.tar.bz2 && cd Vala-version
./configure && make && sudo make install
```

If all goes well and there are no errors, then running *valac --help* in your terminal should display the help text. Now we're ready to play with Vala and Genie.